WHEN A DIVIDED POWER STRUCTURE DESCENDS ALONG A QUOTIENT

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Editorial note. No omission markers in-span; proof complete as written.

Allowed standard prerequisites (editorial).

- Definition of a divided power structure. definition-divided-powers (same chapter, definitional anchor). Axiom (4) drives the well-definedness and the final induction.

Source reference. Original definition 07GL: sources/stacks-dpa/pinned-original-source.tex, source lines 253–253. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let (A, I, γ) be a divided power ring.*

(1) *If $\varphi : (A, I, \gamma) \to (B, J, \delta)$ is a homomorphism of divided power rings, then $\mathrm{Ker}(\varphi) \cap I$ is preserved by γ_n for all $n \geq 1$.*
(2) *Let $\mathfrak{a} \subset A$ be an ideal and set $I' = I \cap \mathfrak{a}$. The following are equivalent*
   (a) *I' is preserved by γ_n for all $n > 0$,*
   (b) *γ extends to $A/\mathfrak{a}$, and*
   (c) *there exist a set of generators x_i of I' as an ideal such that $\gamma_n(x_i) \in I'$ for all $n > 0$.*

Proof. Proof of (1). This is clear. Assume (2)(a). Define $\bar{\gamma}_n(x \bmod I') = \gamma_n(x) \bmod I'$ for $x \in I$. This is well defined since $\gamma_n(x + y) = \gamma_n(x) \bmod I'$ for $y \in I'$ by Definition 07GL (4) and the fact that $\gamma_j(y) \in I'$ by assumption. It is clear that $\bar{\gamma}$ is a divided power structure as γ is one. Hence (2)(b) holds. Also, (2)(b) implies (2)(a) by part (1). It is clear that (2)(a) implies (2)(c). Assume (2)(c). Note that $\gamma_n(x) = a^n \gamma_n(x_i) \in I'$ for $x = ax_i$. Hence we see that $\gamma_n(x) \in I'$ for a set of generators of I' as an abelian group. By induction on the length of an expression in terms of these, it suffices to prove $\forall n : \gamma_n(x + y) \in I'$ if $\forall n : \gamma_n(x), \gamma_n(y) \in I'$. This follows immediately from the fourth axiom of a divided power structure. □

Date: stacks-project:dpa.tex:local-snapshot-20261012; GNU FDL 1.2 or later.